

Problem 1 Laplace transform [10 pts]

- a) Compute the Laplace transform of

$$f(t) = 3 \sin(2t)e^t.$$

- b) Compute the inverse Laplace transform of

$$F(s) = \frac{3s^2 - 11s + 12}{s(s-2)^2}.$$

- c) Use the Laplace transform to solve the initial value problem

$$y'(t) + 3y(t) = \exp(2t), \quad y(0) = 3.$$

Solution

- a) By the table, we know the Laplace transform of $3 \sin(t)$ is

$$\mathcal{L}[3 \sin(2t)](s) = 3\mathcal{L}[\sin(2t)](s) = 3 \frac{2}{s^2 + 2^2} = \frac{6}{s^2 + 4}.$$

Moreover, multiplication by e^{at} gives a shift $s \mapsto s - a$, so

$$\mathcal{L}[f(t)](s) = \mathcal{L}[3 \sin(2t)e^t](s) = \frac{6}{(s-1)^2 + 4} = \frac{6}{s^2 - 2s + 5}$$

- b) We use a partial fraction decomposition

$$\begin{aligned} \frac{3s^2 - 11s + 12}{s(s-2)^2} &= \frac{A}{s} + \frac{B}{s-2} + \frac{C}{(s-2)^2} \\ &= \frac{A(s-2)^2 + Bs(s-2) + Cs}{s(s-2)^2} \\ &= \frac{(A+B)s^2 + (-4A-2B+C)s + 4A}{s(s-2)^2} \end{aligned}$$

Identifying,

$$\begin{cases} 4A = 12 \\ -4A - 2B + C = -11 \\ A + B = 3 \end{cases} \Leftrightarrow \begin{cases} A = 3 \\ B = 0 \\ C = 1 \end{cases}$$

so

$$F(s) = \frac{3}{s} + \frac{1}{(s-2)^2}.$$

By table we find

$$\mathcal{L}^{-1}[F(s)](t) = \mathcal{L}^{-1}\left[\frac{3}{s}\right](t) + \mathcal{L}^{-1}\left[\frac{1}{(s-2)^2}\right](t) = 3 + te^{2t}$$

c) Taking the Laplace transform of the initial value problem gives

$$sY(s) - 3 + 3Y(s) = \frac{1}{s-2}.$$

Solving for $Y(s)$ we obtain

$$Y(s) = \frac{1}{(s-2)(s+3)} + \frac{3}{s+3}.$$

By a partial fraction decomposition of the first term we get

$$Y(s) = \frac{1}{5(s-2)} - \frac{1}{5(s+3)} + \frac{3}{s+3} = \frac{1}{5} \left(\frac{1}{s-2} + \frac{14}{s+3} \right).$$

Taking the inverse Laplace transform, we obtain the solution

$$y(t) = \frac{1}{5}(e^{2t} + 14e^{-3t}).$$

Grading manual

a) (2p)

b) (4p)

c) (4p)

Problem 2 **Fourier series [10 pts]** Extend the function

$$f(x) = 2x,$$

defined on the interval $[0, 2)$, to the interval $[0, 4)$ in such a way that the periodic extension of this function is **even** with a period of 4. Find the Fourier series of this new function, and explicitly write down the first 4 nonvanishing terms of the Fourier series.

Solution The resulting even extension is

$$f_1(x) = \begin{cases} 2x & 0 \leq x < 2 \\ -2x + 8 & 2 \leq x < 4, \end{cases}$$

$f_1(x) = f_1(x + 4n)$ ($n \in \mathbb{Z}$), with period 4. We have

$$\begin{aligned} a_0 &= \frac{1}{2L} \int_{-L}^L f_1(x) dx \\ &= \frac{1}{4} \int_{-2}^2 f_1(x) dx \\ &= \frac{1}{2} \int_0^2 f_1(x) dx \\ &= \frac{1}{2} \int_0^2 2x dx \\ &= \left[\frac{x^2}{2} \right]_0^2 \\ &= 2. \end{aligned}$$

Moreover,

$$\begin{aligned}
 a_n &= \frac{1}{L} \int_{-L}^L \underbrace{f_1(x)}_{\text{even}} \underbrace{\cos\left(\frac{\pi nx}{L}\right)}_{\text{even}} dx \\
 &= \frac{2}{L} \int_0^L f_1(x) \cos\left(\frac{\pi nx}{L}\right) dx \\
 &= \int_0^L 2x \cos\left(\frac{\pi nx}{L}\right) dx \\
 &= \underbrace{\left[2x \frac{L}{\pi n} \sin\left(\frac{\pi nx}{L}\right)\right]_0^L}_{=0} - \int_0^L 2 \frac{L}{\pi n} \sin\left(\frac{\pi nx}{L}\right) dx \\
 &= -\frac{2L}{\pi n} \cdot \frac{L}{\pi n} \left[-\cos\left(\frac{\pi nx}{L}\right)\right]_0^L \\
 &= \frac{2L^2}{\pi^2 n^2} \left(\cos\left(\frac{\pi 2n}{L}\right) - 1\right) \\
 &= \frac{2L^2}{\pi^2 n^2} (\cos(\pi n) - 1).
 \end{aligned}$$

Therefore, we have for

$$\begin{aligned}
 n = 1 : \quad a_1 &= \frac{8}{\pi^2} \cdot (-2) = \frac{-16}{\pi^2} \\
 n = 2 : \quad a_2 &= 0 \\
 n = 3 : \quad a_3 &= \frac{8}{\pi^2 \cdot 3^2} \cdot (-2) = \frac{-16}{9\pi^2} \\
 n = 4 : \quad a_4 &= 0 \\
 n = 5 : \quad a_5 &= \frac{8}{\pi^2 \cdot 5^2} \cdot (-2) = \frac{-16}{25\pi^2}.
 \end{aligned}$$

The resulting Fourier series is

$$\begin{aligned}
 f_1(x) &= 2 + \sum_{n=1}^{\infty} \frac{2L^2}{\pi^2 n^2} (\cos(\pi n) - 1) \cos\left(\frac{n\pi x}{2}\right) \\
 &\approx 2 - \frac{16}{\pi^2} \cos\left(\frac{\pi x}{2}\right) - \frac{16}{9\pi^2} \cos\left(\frac{3\pi x}{2}\right) - \frac{16}{25\pi^2} \cos\left(\frac{5\pi x}{2}\right) - \dots
 \end{aligned}$$

Grading manual (1p) for the proper even extension

(2p) for a_0

(3p) for a_n

(3p) for a_1, a_3, a_5

(1p) for the Fourier series.

Problem 3 Numerical methods for ODEs [10 pts] Consider python code that solves an initial value problem with a Runge-Kutta method and prints the resulting x and y value after N steps:

```

1 def f(x, y):
2     return x * y**2
3
4 def step(x, y, h):
5     k1 = f(x, y)
6     k2 = f(x + h, y + h * k1)
7     y_new = y + (h / 2) * (k1 + k2)
8     x_new = x + h
9     return x_new, y_new
10
11 x = 0
12 y = 0.5
13 h = 0.1
14 N = 20
15
16 for n in range(N):
17     x, y = step(x, y, h)
18
19 print(x, y)

```

- What is the initial value problem that the code solves?
- Write the Butcher tableau for the method used. Is the method explicit or implicit? motivate your answer.
- Compute one step of the algorithm in the code by hand. That is, what is the output if we run the code with $N = 1$ instead?

Solution

- We find the ODE in the initial value problem from the function f and the initial condition is given by the values that x and y are set to before the for loop. This gives the IVP

$$y'(x) = xy^2, \quad y(0) = 0.5.$$

- The method used is Heun's method with Butcher tableau

$$\begin{array}{c|cc} 0 & 0 & 0 \\ 1 & 1 & 0 \\ \hline & 1/2 & 1/2 \end{array}$$

The method is explicit because $a_{ij} = 0$ whenever $j \geq i$.

c) The step size is given by $h = 0.1$, $x_0 = 0$ and $y_0 = 0.5$ so we compute

$$\begin{aligned} k_1 &= f(x_0, y_0) = x_0 y_0^2 = 0 \times 0.5^2 = 0, \\ k_2 &= f(x_0 + h, y_0 + h k_1) = (x_0 + h)(y_0 + h k_1)^2 \\ &= (0 + 0.1)(0.5 + 0.1 \times 0)^2 = 0.1 \times 0.5^2 = 0.1 \times 0.25 = 0.025 \end{aligned}$$

giving

$$y_1 = y_0 + \frac{h}{2}(k_1 + k_2) = 0.5 + 0.05 \times (0.025) = 0.5 + 0.00125 = 0.50125$$

and

$$x_1 = x_0 + h = 0 + 0.1 = 0.1.$$

With $N = 1$ the code prints 0.1 and 0.50125.

Grading manual

a) (2p)

b) (3p)

c) (5p)

Problem 4 Interpolation [10 pts] Consider the data points

$$\begin{array}{c|ccc} x_i & 0 & 2 & 4 \\ \hline y_i = f(x_i) & 1 & 0 & 1 \end{array}$$

- a) Use Lagrange interpolation to find the polynomial of minimal degree interpolating these points. Express the polynomial in the form

$$p_n(x) = a_n x^n + \cdots + a_1 x + a_0.$$

- b) Determine the Newton form of the interpolating polynomial.

- c) Verify that the solutions in (b) and (c) are the same.

I suppose that this should be "Verify that the solutions in (a) and (b) are the same."

- d) Use your result to find an approximation to $f(1)$.

Solution.

- a) We calculate the cardinal functions:

$$\begin{aligned} \ell_0 &= \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = \frac{(x - 2)(x - 4)}{(-2)(-4)} = \frac{x^2 - 6x + 8}{8} \\ \ell_1 &= \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} = \frac{x(x - 4)}{2 \cdot 2} = \frac{x^2 - 4x}{4} \\ \ell_2 &= \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} = \frac{x(x - 2)}{4 \cdot 2} = \frac{x^2 - 2x + 8}{8} \end{aligned}$$

Note that ℓ_1 does not need to be computed, as $y_1 = 0$. We get

$$p_2(x) = \underbrace{y_0}_{=1} \cdot \ell_0 + \underbrace{y_1}_{=0} \cdot \ell_1 + \underbrace{y_2}_{=1} \cdot \ell_2 = \frac{x^2 - 6x + 8}{8} + \frac{x^2 - 2x + 8}{8} = \frac{1}{4}x^2 - x + 1.$$

- b) We have

$$\begin{aligned} f[x_0, x_1] &= \frac{f(x_1) - f(x_0)}{x_1 - x_0} = -\frac{1}{2} \\ f[x_1, x_2] &= \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{1}{2} \\ f[x_0, x_1, x_2] &= \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = \frac{1}{4}. \end{aligned}$$

The table reads

x_i	y_i
0	1
2	0
4	1

The interpolating polynomial becomes

$$p_2(x) = f[x_0, x_1, x_2] \cdot (x - x_0) \cdot (x - x_1) + f[x_0, x_1] \cdot (x - x_0) + f[x_0] = \frac{1}{4}x(x - 2) - \frac{1}{2}x + 1.$$

c) We have

$$p_2(x) = \frac{1}{4}x(x - 2) - \frac{1}{2}x + 1 = \frac{1}{4}(x^2 - 2x) - \frac{1}{2}x + 1 = \frac{1}{4}x^2 - x + 1.$$

d) An approximation to $f(1)$ is

$$p(1) = \frac{1}{4} - \frac{1}{2} - \frac{1}{2} + 1 = \frac{1}{4}.$$

Grading manual

a) (4p)

b) (4p)

c) (1p)

d) (1p)

Problem 5 Heat Equation [10 pts] Solve the problem

$$\left\{ \begin{array}{ll} \frac{\partial}{\partial t} u(t, x) = 9 \frac{\partial^2}{\partial x^2} u(t, x), & \text{for } 0 \leq t, 0 \leq x \leq \pi, \\ \frac{\partial}{\partial x} u(t, 0) = \frac{\partial}{\partial x} u(t, \pi) = 0 & \\ \text{correction: } = \frac{\partial}{\partial x} u(t, \pi) = 0, & \text{for } 0 \leq t, \\ u(0, x) = 3 \cos(x) - \cos(3x), & \text{for } 0 \leq x \leq \pi, \end{array} \right.$$

using separation of variables.

Solution

- a) We start by separation of variables setting $u(t, x) = F(x)G(t)$. Substituting this into the equation itself and dividing by $9F(x)G(t)$ gives

$$\frac{F''(x)}{F(x)} = \frac{G'(t)}{9G(t)} = k,$$

where the left hand side is independent of t and the right-hand side is independent of x , so they have to equal the same constant k . In other words we get the equations

$$F''(x) - kF(x) = 0,$$

and

$$G'(t) - 9kG(t) = 0.$$

- b) For $u(t, x)$ to satisfy the boundary conditions (and not be $u(t, x) \equiv 0$), we need $F'(0) = F'(\pi) = 0$. The general solution to the equation for F depends on the sign of k and we get

$$\begin{aligned} k = p^2 > 0 & \quad \text{gives } F(x) = Ae^{px} + Be^{-px}, \\ k = 0 & \quad \text{gives } F(x) = Ax + B, \\ k = -p^2 < 0 & \quad \text{gives } F(x) = A \sin(px) + B \cos(px). \end{aligned}$$

In the first case the boundary conditions are satisfied only for $A = B = 0$ and in the second case they are satisfied for $A = 0$ and for an arbitrary B . In the third case we can differentiate the expression to obtain $F'(x) = pA \cos(px) - pB \sin(px)$, so the conditions become

$$0 = F'(0) = pA$$

which gives $A = 0$ because $p^2 \neq 0$ and (with $A = 0$)

$$0 = F'(\pi) = -pB \sin(p\pi),$$

which gives $B = 0$ unless $p = n \in \mathbb{Z}$, then it holds for an arbitrary B . So we get the nonzero solutions

$$F_n(x) = B_n \cos(nx), \quad n \in \mathbb{Z}.$$

c) The corresponding G_n then solve $G'(t) + (3n)^2 G(t) = 0$, which has general solution

$$G_n(t) = C e^{-(3n)^2 t} = 0, \quad n \in \mathbb{Z}.$$

d) Since we already have an arbitrary constant in F_n we can set $C = 1$ and we obtain the solutions

$$u_n(t, x) = F_n(x) G_n(t) = B_n \cos(nx) e^{-(3n)^2 t}, \quad n \in \mathbb{Z}.$$

Due to linearity and homogeneity, the linear combination of two solutions is still a solution, so to obtain a solution that satisfies the initial condition we can sum u_1 with $B_1 = 3$ and u_3 with $B_3 = -1$. Thus the solution to the problem is

$$u(t, x) = 3 \cos(x) e^{-9t} - \cos(3x) e^{-81t}.$$

Grading manual

a) (3p)

b) (4p)

c) (1p)

d) (2p)

Problem 6 Wave Equation [10 pts]

- a) Verify by explicit computations that the function:

$$\hat{u}(x, t) = \sin\left(\frac{m\pi}{L}x\right) \cos\left(c\frac{m\pi}{L}t\right)$$

and the function

$$\tilde{u}(x, t) = \frac{1}{2} \left(\sin\left(\frac{m\pi}{L}(x - ct)\right) + \sin\left(\frac{m\pi}{L}(x + ct)\right) \right)$$

are both solutions of the 1-D wave equation

$$u_{tt} = c^2 u_{xx},$$

with initial conditions conditions (initial shape) $u(x, 0) = f(x) = \sin(\frac{m\pi}{L}x)$ and (vibration from rest) $u_t(x, 0) = g(x) = 0$.

- b) Show that
- $\hat{u} = \tilde{u}$
- .

Hint: Use a trigonometric identity.

- c) Suppose that the functions
- f
- and
- g
- have the following symmetry properties for
- $0 \leq x \leq L$
- :
- $f(L - x) = f(x)$
- and
- $g(L - x) = g(x)$
- , i.e., both are symmetric around
- $2x = L$
- . Use d'Alembert's solution of the 1-D wave equation

$$u(x, t) = \frac{1}{2} \left(f^*(x + ct) + f^*(x - ct) + \frac{1}{c} \int_{x-ct}^{x+ct} g^*(s) ds \right),$$

where f^* and g^* are the (2L-periodic) odd extensions of f respectively g , to show for all $0 < x < L$ and $t > 0$ that

$$u(x, t + \frac{L}{c}) = -u(x, t).$$

Solution

- a)

$$\hat{u}_t = -c \frac{m\pi}{L} \sin\left(\frac{m\pi}{L}x\right) \sin\left(c\frac{m\pi}{L}t\right)$$

$$\hat{u}_{tt} = -c^2 \frac{m^2 \pi^2}{L^2} \sin\left(\frac{m\pi}{L}x\right) \cos\left(c\frac{m\pi}{L}t\right)$$

$$\hat{u}_x = \frac{m\pi}{L} \cos\left(\frac{m\pi}{L}x\right) \cos\left(c\frac{m\pi}{L}t\right)$$

$$\hat{u}_{xx} = -\frac{m^2 \pi^2}{L^2} \sin\left(\frac{m\pi}{L}x\right) \cos\left(c\frac{m\pi}{L}t\right)$$

We see that $\hat{u}_{tt} = c^2 \hat{u}_{xx}$.

$$\begin{aligned}\tilde{u}_t &= \frac{1}{2} \left(-c \frac{m\pi}{L} \cos\left(\frac{m\pi}{L}(x-ct)\right) + c \frac{m\pi}{L} \cos\left(\frac{m\pi}{L}(x+ct)\right) \right) \\ \tilde{u}_{tt} &= \frac{1}{2} \left(-c^2 \frac{m^2\pi^2}{L^2} \sin\left(\frac{m\pi}{L}(x-ct)\right) - c^2 \frac{m^2\pi^2}{L^2} \sin\left(\frac{m\pi}{L}(x+ct)\right) \right) \\ \tilde{u}_x &= \frac{1}{2} \left(\frac{m\pi}{L} \cos\left(\frac{m\pi}{L}(x-ct)\right) + \frac{m\pi}{L} \cos\left(\frac{m\pi}{L}(x+ct)\right) \right) \\ \tilde{u}_{xx} &= \frac{1}{2} \left(-\frac{m^2\pi^2}{L^2} \sin\left(\frac{m\pi}{L}(x-ct)\right) - \frac{m^2\pi^2}{L^2} \sin\left(\frac{m\pi}{L}(x+ct)\right) \right)\end{aligned}$$

We see that $\tilde{u}_{tt} = c^2 \tilde{u}_{xx}$.

b) Use trigonometric identity

$$2 \sin(x) \cos(y) = \sin(x+y) + \sin(x-y).$$

c) Recall D'Alembert's solution of the 1-D wave equation

$$u(x, t) = \frac{1}{2} \left(f^*(x+ct) + f^*(x-ct) + \frac{1}{c} \int_{x-ct}^{x+ct} g^*(s) ds \right).$$

Here f^* and g^* are odd and $2L$ -periodic. Then

c1)

$$u(x, t + \frac{L}{c}) = \frac{1}{2} \left(f^*(x+ct+L) + f^*(x-ct-L) + \frac{1}{c} \int_{x-ct-L}^{x+ct+L} g^*(s) ds \right).$$

Properties of f^* and g^* imply:

c2) (1p)

$$f^*(x+ct+L) = f^*(x+ct+L-2L) = -f^*(L-x-ct) = -f^*(x+ct)$$

and

c2) (1p)

$$f^*(x-ct-L) = -f^*(L-x+ct) = -f^*(L-(x-ct)) = -f^*(x-ct)$$

and

c2) (1p)

$$g^*(s+L) = -g^*(-s-L) = -g^*(-s-L+2L) = -g^*(L-s) = -g^*(s).$$

A change of variables then gives

$$\begin{aligned} \frac{1}{c} \int_{x-ct-L}^{x+ct+L} g^*(s) ds &\stackrel{c3)}{=} \frac{1}{c} \int_{x-ct}^{x+ct} g^*(s+L) ds \\ &\stackrel{c4)}{=} \frac{-1}{c} \int_{x-ct}^{x+ct} g^*(s) ds. \end{aligned}$$

Adding these identities up yields the statement

c5)

$$u(x, t + \frac{L}{c}) = -u(x, t).$$

Grading manual

a) (1p)

b) (2p)

c) (7p):

c1) (1p), c2) (1p)+(1p)+(1p), c3) (1p), c4) (1p), c5) (1p)

Problem 7 Fourier Transform [10 pts]

Suppose the function f is integrable and even. Consider the function $g(t) = \cos(at)$. Show that for all $a \in \mathbb{R}$, we have that the convolution product

$$(f * g)(t) = \cos(at)\hat{f}(a).$$

Recall that $\hat{f}(a) = \mathcal{F}(f)(a)$ is the Fourier transform of f evaluated at a .

Hint: You may use the trigonometric identity $\cos(x - y) = \cos(x)\cos(y) + \sin(x)\sin(y)$.

For Mathematics 4D: there should be an extra factor $\sqrt{2\pi}$ in the answer (i.e. they should show $(f * g)(t) = \sqrt{2\pi}\cos(at)\hat{f}(a)$) because the convolution was defined without $\frac{1}{\sqrt{2\pi}}$ in the lectures.

Solution a) We compute

$$(f * g)(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) \cos(a(t - x)) dx \quad (1)$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) (\cos(at)\cos(ax) + \sin(at)\sin(ax)) dx. \quad (2)$$

b) Since the function f is even ($f(-t) = f(t)$, for all t), we have that the product $f(t)\sin(at)$ is an odd function, and therefore the integral $\int_{-\infty}^{\infty} f(x)\sin(ax)dx = 0$.

c) We continue the computation of $(f * g)(t)$:

$$(f * g)(t) \stackrel{c1)}{=} \cos(at) \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) \cos(ax) dx \quad (3)$$

$$\stackrel{c2)}{=} \cos(at) \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) (\cos(ax) - i \sin(ax)) dx \quad (4)$$

$$\stackrel{c3)}{=} \cos(at) \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) \exp(-iax) dx \quad (5)$$

$$\stackrel{c4)}{=} \cos(at) \hat{f}(a). \quad (6)$$

Grading manual (10p)

a) (1p)

b) (2p)

c) (7p):

c1) (1p), c2) (3p), c3) (2p), c4) (1p)

Problem 8 Numerical integration [10 pts]

- a) Find the values of w_j , $j = -1, 0, 1, 2$, such that the quadrature rule

$$Q[f] = w_{-1}f(-1) + w_0f(0) + w_1f(1) + w_2f(2)$$

gives the correct value for the integral

$$\int_0^1 f(s)ds,$$

when f is any polynomial of degree three.

- b) Denote by T_1 the trapezoid rule using a single subinterval. Denote by M_1 the midpoint rule using a single subinterval. Denote By S_2 the Simpson's rule using a single subinterval. Show that, for any continuous function f , and any interval $[a, b]$

$$S_2[f](a, b) = \frac{1}{3}T_1[f](a, b) + \frac{2}{3}M_1[f](a, b).$$

- c) c1) Show that the quadratic Legendre polynomial $P_2(x) = \frac{1}{2}(3x^2 - 1)$ is orthogonal over $[-1, 1]$ to all linear polynomials.
c2) The first two Legendre polynomials are

$$P_0(x) = 1, \quad P_1(x) = x.$$

It can be shown that the other Legendre polynomials, P_{n+1} , $n > 1$, satisfy the recurrence relation

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x).$$

Use this to show by induction that the leading coefficient for the Legendre polynomials is

$$k_n = \frac{(2n)!}{2^n(n!)^2}$$

Solution

- a) To compute the numbers w_j , $j = -1, 0, 1, 2$, we demand that the quadrature rule integrates the polynomials $1, x, x^2, x^3$ exactly. Then it will integrate any polynomial of order three exactly. We arrive at the system

$$\begin{aligned} w_{-1} + w_0 + w_1 + w_2 &= 1 \\ -w_{-1} + w_1 + 2w_2 &= \frac{1}{2} \\ w_{-1} + w_1 + 4w_2 &= \frac{1}{3} \\ -w_{-1} + w_1 + 8w_2 &= \frac{1}{4} \end{aligned}$$

Solving this linear system yields $w_{-1} = w_2 = -\frac{1}{24}$ and $w_0 = w_1 = \frac{13}{24}$.

- b) We denote by c the midpoint of the interval $[a, b]$. Then we have

$$T_1[f](a, b) = \frac{(b-a)}{2}(f(b) + f(a))$$

and

$$M_1[f](a, b) = (b-a)f(c).$$

Then a direct computation shows

$$\begin{aligned} \frac{1}{3}T_1[f](a, b) + \frac{2}{2}M_1[f](a, b) &= \frac{1}{6}(b-a)(f(b) + f(a)) + \frac{2}{3}(b-a)f(c) \\ &= \frac{\frac{(b-a)}{2}}{3}(f(a) + 4f(c) + f(b)) = S_2[f](a, b). \end{aligned}$$

- c) c1) Take a general linear polynomial $q(x) = Ax + B$. Then

$$\begin{aligned} \int_{-1}^1 P_2(s)q(s)ds &= \frac{1}{2} \int_{-1}^1 3As^3 + 3Bs^2 - As - Bds \\ &= \frac{1}{2} \left(\frac{3}{4}As^4 + Bs^3 - \frac{1}{2}As^2 - Bs \right) \Big|_{-1}^1 \\ &= \frac{3}{4}A + B - \frac{1}{2}A - B - \frac{3}{4}A + B + \frac{1}{2}A - B \\ &= 0 \end{aligned}$$

- c2) The statement holds for P_0 , P_1 , and P_2 . Now we use the recurrence relation on the leading order coefficient k_{n+1} in the Legendre polynomial P_{n+1} , that is:

$$(n+1)k_{n+1} = (2n+1)k_n.$$

From this together with the induction hypotheses, we obtain:

$$\begin{aligned}k_{n+1} &= \frac{(2n+1)}{(n+1)} \frac{(2n)!}{2^n (n!)^2} \\&= \frac{(2n+2)(2n+1)}{2(n+1)(n+1)} \frac{(2n)!}{2^n (n!)^2} \\&= \frac{(2n+2)!}{2^{n+1} ((n+1)!)^2}.\end{aligned}$$

Grading manual (10p)

a) (2p)

b) (2p)

c) (6p)

c1) (2p), c2) (4p)

Problem 9 Numerics for Nonlinear Equations [10 pts] Let r be a solution (root) of the following equation

$$x - \cos(3x) = 0, \quad 0 \leq x \leq 1.$$

- a) Show that a root exists by using the intermediate value theorem, and show that the solution is unique.
- b) Perform two iterations of the bisection method on the interval $[0, 1]$. What is an upper bound of the error after these 2 iterations?
- c) How many iterations are necessary in order to compute the root up to the tolerance $tol = 0.01$?

Solution

- a) The function $f(x) = x - \cos(3x)$ is continuous on the interval $[0, 1]$. We have $f(0) = 0 - \cos(0) = -1$ and $f(1) = 1 - \cos(3) \approx 1 - 0.98999 = 1.98999$. Therefore, by the intermediate value theorem, there exists a zero in $[0, 1]$. This solution is unique as

$$f'(x) = 1 + 3 \sin(3x) \geq 0$$

for all $x \in [0, 1]$.

- b) $c_0 = \frac{0+1}{2} = \frac{1}{2}$, $f(c_0) = \frac{1}{2} - \cos(3/2) \approx 0.429 > 0$. So, $a_1 = 0$ and $b_1 = 0.429$. We get $c_1 = \frac{0+1/2}{2} = 1/4$. $f(c_1) = 1/4 - \cos(3/4) \approx -0.482$. So, $a_2 = 1/4$ and $b_2 = 1/2$, and the approximation of the root after 2 iterations is $c_2 = \frac{1/2+1/4}{2} = 3/8$. The error satisfies

$$|c_2 - r| \leq \frac{b_2 - a_2}{2} = \frac{1/4}{2} = 1/8.$$

- c) We use the formula

$$\frac{\log\left(\frac{b-a}{2 \cdot tol}\right)}{\log 2} = \frac{\log\left(\frac{1}{2 \cdot 0.01}\right)}{\log 2} = 5.64 \leq k.$$

This means that we need at most 6 iterations until we get an approximation to the root that is within the given tolerance.

Grading manual

a) (4p)

b) (4p)

c) (2p)

Grading Scale (The following grading scheme is adjusted from this one:
<https://i.ntnu.no/wiki/-/wiki/English/Grading+scale+using+percentage+points>):

A	81–90
B	70–80
C	60–69
D	49–59
E	38–48
F	0–37

Because of the two errors in questions 4c and 5, we reduced the grading scale by 8 points (1 from 4c and 7 points from 5), such that the following scale is to be used:

A	73–90
B	62–72
C	52–61
D	41–51
E	30–40
F	0–29